

ISHITA DHOPESHWAR

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https://scholar.google.com/citations?view_op=list_works&hl=en&hl=en&user=_UKdFPAAAAAJ
<https://patents.google.com/patent/US12073277B2/en>

EDUCATION

University at Buffalo, The State University of New York, USA , Master of Science, Electrical Engineering
University of Pune, PVG, India, Bachelor of Engineering, Electronics and Telecommunication Engineering

January-2023
June-2019

WORK EXPERIENCE

Research Scholar II- Tarana Wireless

January 2023- Present

- Drove end-to-end product development—from feature definition to deployment strategy—resulting in a successful product launch.
- Evaluated software-hardware interoperability, improving device performance by 5% while addressing significant field-impacting issues.
- Enhanced the Remote Node procedure for network entry, advancing algorithm efficiency in Radio Node Network Entry while reducing device search time by ½ and identifying bugs in RACH handshake. Additionally identifying issues with the primary BN search timer.
- Planned RF layout based on azimuth for a link using python to detect if the remote node is connected to its optimal base node. This improved installer efficiency by advising install azimuth for the remote node..
- Led, developed and presented lab setup utilizing new technology improving traffic capabilities, reachability and debuggability.
- Enhanced new wireless spectrum adoption and supported customer pilots for link improvement under unsuitable weather conditions
- Led and managed new features for live feedback on system debuggability with enhanced UI for elevated user experience.
- Authored confluence articles and internal app notes for knowledge transfer and documentation.
- Validated new hardware and software for the hardware-early-access-program(HEAP) through various lab and OTA setups.
- Involved in early stages of product development supporting both firmware and hardware validation

Research Assistant- [Wireless Intelligent Networking and Security Laboratory](#), University at Buffalo

January 2021- January 2023

- Project lead: RFID development in collaboration with ARMOR IIMAK (UB-ARMOR IIMAK project, Chipless RFID development, mmWave communications
- Invented state-of-the-art RFID printable tag antenna (patented) with 2x higher read range compared to commercial tags with low cost and time efficient manufacturing techniques.
- Circuit and Algorithm Design for Chipless Wireless Sensing and Communications to reduce the construction time further and minimize the need for chip based RFID systems.
- Designed a testbed for Software-Defined mmWave Communications to establish a 1ft long wireless link while evaluating the setup for a mobile base station. Prototyping mmWave radar using M-cube: mmWave router developed by UCSD to set-up reconfigurable intelligent surfaces and phased array control.
- Prototypes OFDM using Orthogonality in AM, Chirp and FM signals, comparing spectral efficiency, number of channels, guard band allocation, and channel characteristics interpreting the increase in number of channels and less guard bands for chirp signals compared to traditional sin based OFDM channels.

Summer Intern - Vodafone

May 2017-July

2017

Project: Data Distribution of active cell phone tower in Home Location Register

- Algorithm design to automatically document active cell users in Vodafone Mumbai Home Location Register using Excel Macros
- LTE Planning and analysis
- Capex and Opex for cost calculation and inventory management

Research Assistant- Pune Vidhyarthi Griha's College of Engineering and technology

August 2018-May 2019

- Project Lead Autonomous multiutility "Follow me" Robot-
- Formulated a process to compare on-board GPS coordinates with the coordinates streamed by the user's smartphone.
- Calculated the heading and bearing to which the robot will identify its location and direct itself towards the smartphone.

PUBLICATIONS & INTELLECTUAL PROPERTY

- Ishita Dhopeshwar, M. McManus, D. Harrison, B. Ralston, C. Janson, A. Rae, A. Levesque, and Z. Guan, "Antenna Design and Measurements for Conductive Thermal Transfer Printing Based RFID Production," in **IEEE Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)**, Toronto, Canada, 5-8 September, 2023.
- **US 12,073,277 B2 Granted, FR patent granted FR3145240B1 and China patent application CN118673951A** : "Antenna Design and Measurements for Conductive Thermal Transfer Printing Based RFID Production" as inventor
- Ishita Dhopeshwar, "Antenna Design and Measurements for Conductive Thermal Transfer Printing Based RFID Production"-thesis dissertation

CONFERENCES & AWARDS

- Grace Hopper Conference and Scholar March 2021
- Reviewed papers for peer-reviewed IEEE conferences
- Grace Hopper Conference Mentor March 2022
- GradSWE speaker and Secretary at University at Buffalo February 2021-December 2022
- Scholarship from National Science Foundation (NSF) for Young Gladiators May 2022
- Young Gladiators- A workshop organized in collaboration with Northeastern University for hands on experience on Colosseum, world's largest RF emulator June 2015-June 2018
- Technical head and Joint General Secretary for Telecommunications Engineering Student Association